Carbohydrate Polymers 272 (2021) 118473

Contents lists available at ScienceDirect

Carbohydrate Polymers

journal homepage: www.elsevier.com/locate/carbpol

Room temperature dissolving cellulose with a metal salt hydrate-based deep eutectic solvent

Zhihan Tong, Juan Meng, Shi Liu, Yongzhuang Liu, Suqing Zeng, Lei Wang, Qinqin Xia *, Haipeng Yu *

Key Laboratory of Bio-based Material Science and Technology of Ministry of Education, Northeast Forestry University, Harbin 150040, Heilongjiang, China

ARTICLE INFO

Keywords:
Aqueous solutions
Biorefining
Cellulose
Deep eutectic solvents
DFT calculations
Solubility
Sustainable chemistry

ABSTRACT

Abundant and renewable cellulose is a potential candidate for petroleum-derived synthetic polymers. However, the efficient dissolution of this material is problematic because of the high cost, severe reaction condition (e.g., high temperature) and environmentally unfriendly (e.g., toxic reagents, and solvent recyclability). Herein, to realize the room temperature dissolution of cellulose with an inexpensive and eco-friendly solvent, we design a novel low-cost deep eutectic solvent that is composed of zinc chloride, water and phosphoric acid for the efficient dissolution of cellulose. This solvent is featured as having both the superior hydrogen bonding acidity and the hydrogen bonding basicity, and thus can act as a hydrogen bond molecular scissors to cleave the hydrogen bonds within cellulose. In this process, microcrystalline cellulose can be easily dissolved in the solvent at room temperature with a dissolution ratio up to 15 wt%. The dissolved cellulose can also be recovered without any derivatization. The universality, recyclability and pilot production of dissolving cellulose using this solvent are also demonstrated. This work provides a new strategy for the design of novel deep eutectic solvent capable of disrupting the hydrogen bonds of cellulose under mild conditions.

1. Introduction

As a readily available, naturally occurring biopolymer, cellulose can be considered an almost inexhaustible resource based on the annual growth of 1.5×10^{12} tons of this substance (Klemm et al., 2005; Ragauskas et al., 2006). The dissolution process is to tune the structure of cellulose molecules from the microscope, and construct a series of new types of fibers, hydrogels, membranes and artificial silk materials (Jia et al., 2019; Pang et al., 2021; Tardy et al., 2021), opening up a way to efficiently utilize cellulose resources (Zhu et al., 2021). Cellulose is typically bound to hemicellulose and lignin within the cell walls of plants (Berglund and Burgert, 2018). The robust semicrystalline structure of cellulose via extensive hydrogen bonds (H-bonds) renders it insoluble in water and many organic solvents, which renders the subsequent processes difficult (Martinez, 2016). For these reasons, there is significant interest in the development of new, efficient and environmentally friendly solvents for the dissolution of cellulose.

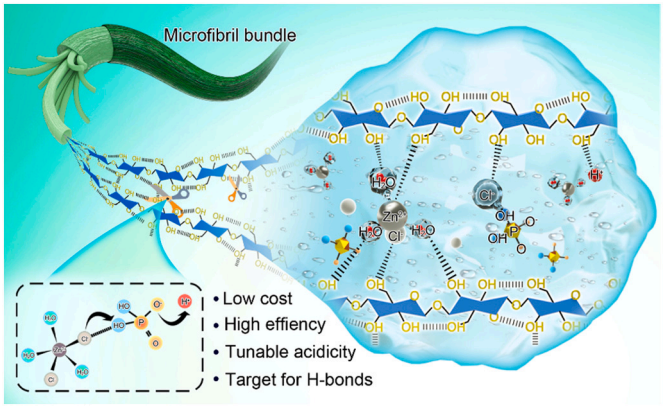
Currently, both the derivatizing and non-derivatizing solvents are commonly used for cellulose dissolution (Yang et al., 2021). Derivatizing solvents (e.g., copper ammonia solution, cadmium/ethylenediamine solution, $NaOH/CS_2$ and carbamate solvent) usually cause significant degradation of the cellulose structure, and cannot be efficiently recycled (Van de Vyver et al., 2011). The development of highly effective, non-derivatizing solvents for cellulose dissolution has thus been of interest for many years. Ionic liquids as the non-derivatizing solvents (De Oliveira and Rinaldi, 2015; Swatloski et al., 2002) are typically composed of large organic cations and low charge density anions (Zhao et al., 2013). During the cellulose dissolution process, the anions can form strong H-bonds with the hydroxyl groups in cellulose chains while the cations primarily provide steric hindrance (Zhang et al., 2016). Ionic liquids are found to readily dissolve cellulose (Ferreira et al., 2019; Lethesh et al., 2020; Wang et al., 2012) but are still costly and difficult to recycle, which limit their industrial applications.

Various ionic liquid-like systems capable of dissolving cellulose at

Abbreviations: AIM, atoms in molecules; BCP, bond critical point; DES, deep eutectic solvent; DFT, density functional theory; H-bond, hydrogen bond; HBA, hydrogen bond acceptor; HBD, hydrogen bond donor; K–T, Kamlet–Taft; MCC, microcrystalline cellulose; RMCC, regenerated microcrystalline cellulose; DP, degree of polymerization.

* Corresponding authors.
E-mail addresses: 2018xiaqinqin@nefu.edu.cn (Q. Xia), yuhaipeng20000@nefu.edu.cn (H. Yu).

https://doi.org/10.1016/j.carbpol.2021.118473
Received 20 April 2021; Received in revised form 19 July 2021; Accepted 19 July 2021
Available online 22 July 2021
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Scheme 1. The conceive of a metal salt hydrate-based deep eutectic solvent (DES) for the room temperature dissolution of cellulose. The structure-activity design of a metal salt hydrate and polyacid forms a special DES capable of acting as the hydrogen bond molecular scissors solvent targeted for breaking hydrogen bonds and thereby dissolving cellulose.

room temperature have been reported, based on early work using alkali/urea aqueous system (Cai and Zhang, 2005). These include tetrabutylammonium hydroxide (TBAH), tetrabutylphosphonium hydroxide (TBPH) (Nguyen et al., 2019), tetrabutylammonium acetate/dimethyl sulfoxide (TBAA/DMSO) (Huang et al., 2016), TBAH/DMSO (Chen et al., 2018), TBAH/urea, and others. These compounds are similar, in that they contain superbase cations that can attack the reducing ends of cellulose chains to initiate free radical reactions, thus disrupting the ordered structure of the cellulose. Subsequent to this, electron donor-acceptor coordination reactions can proceed (Li et al., 2020). The solvation effect of small molecules promotes the dissociation of free ions and facilitates ion mobility, while urea adjusts the amphiphilicity of the system, thereby improving the solubility of cellulose. However, the toxicity issue of these compounds tends to limit their widespread applications.

Alternatively, inorganic molten salt hydrates are reported with similar characteristics to ionic liquids and are of special interest due to their ease of preparation and low cost (Sen et al., 2013). As an example, zinc chloride hydrate ($ZnCl_2{\cdot}nH_2O$) can be used to directly process cellulose within the formula range $2 < n < 4$ (Bodachivskyi et al., 2019; Wilcox et al., 2014). Its similar compound formula (which can also be considered to exist as $[Zn(OH_2)_6]^{2+}[ZnCl_4]^{2-}$) can dissolve cellulose. However, the solubility of cellulose in such inorganic salt hydrates is very low (<5 wt%) and needs high temperature (>80 °C), causing energy consumption and partially hydrolysis. Therefore, it is critical to design a facile and environmentally friendly solvent to effectively dissolve cellulose under mild conditions while at the same time reducing the excessive hydrolysis, thus requiring a unique balance between dissolution and hydrolysis.

A metal salt hydrate-based deep eutectic solvent (DES IV) that involves metal salt hydrate (as H-bond acceptor) in combination with various H-bond donors (e.g., acid, urea, amide) is a promising alternative solvent for dissolving cellulose due to its low cost and high H-bond accepting ability (Chen and Mu, 2019; Jiang et al., 2019; Meng et al., 2021; Sirviö, 2019; Sirviö and Heiskanen, 2017; Wang et al., 2020; Yu et al., 2018, 2019). Here, we intend to design such a type of solvent that can act as "Hydrogen bond molecular scissors" (Luo et al., 2020) to realize the room temperature dissolution of cellulose. Based on the concept of DES IV, a combination of metal salt hydrates and polyacids is hypothesized under certain conditions to improve the dissolution capacity of the mixture (Scheme 1). Water acts as an essential constituent to regulate the behavior of the ions and the catalytic effect. The detrimental effect of excess hydration on the solvent effect is a concern. For this reason, density functional theory (DFT) calculations are employed to study the internal supramolecular interactions among the constituent species at the atomic level. The goal was to obtain an in-depth understanding of the solvation mechanism, combined with an experimental optimization to determine the best performing room temperature cellulose solvent.

2. Experimental

2.1. Cellulose feedstocks

Four types of cellulose were selected as the feedstocks for the dissolution experiments. Microcrystalline cellulose (MCC, 25 μm) was purchased from the Aladdin Reagent Co., Ltd. (Shanghai, China), wood cellulose was extracted from poplar wood using the sodium chlorite method (see Supplementary Note S1), bleached softwood pulp was purchased from the Jiali Trade Co. Ltd. (Liaoning, China) and commercial degreasing cotton was obtained from the Yanggu JYG Hygiene & Health Factory (Shandong, China). All these cellulose feedstocks were characterized with regard to degree of polymerization (DP), average molecular weight, α-cellulose content and crystallinity.

2.2. Preparation of the DESs

Zinc chloride and phosphoric acid (purity 85%) were purchased from the Kermel Chemical Reagent Co., Ltd. (Tianjin, China). Zinc chloride, water and phosphoric acid were mixed at molar ratios of 1:*x*:*y* ($x = 2.5$, 3, 3.5 or 4; $y = 0.2$, 0.4, 0.6, 0.8 or 1.0) and stirred at room temperature (23–26 °C, 40%RH) for 5 min. The resulting solvents are referred to herein as MSA-*x*-*y*.

2.3. Dissolution of cellulose

Trials involving the room temperature dissolution of cellulose were initiated by adding a specific mass of cellulose (5–17 wt%) to a 100 mL round-bottomed flask together with 20 g of the solvent, followed by stirring at 600 rpm. Samples were taken at intervals of 10 min to observe the dimensions (e.g., length, width) and shapes (e.g., fiber or particle) of cellulose using a polarized light microscopy (Smart-POL, Chongqing Optec Instrument Co. Ltd., Chongqing, China). The dissolution time can be considered as the useful information translated into dissolution efficiency.

2.4. DFT calculations

Based on the role of H-bond donor-acceptor pairing in DES, three metal salt hydrates were evaluated for the synthesis of new solvents. $LiCl{\cdot}3H_2O$, $MgCl_2{\cdot}3H_2O$, $ZnCl_2{\cdot}3H_2O$ are denoted as MS1, MS2, MS3, respectively. The structure of cellobiose and molecular interactions between the solvent and cellobiose in this work were studied using the Gaussian 16 package, with all samples optimized at the M06-2X/6-311+G** level. Frequency analyses were carried out for the optimized structures to verify energy minima and provide zero-point energy corrections. Thermochemical characteristics were also assessed to ensure consistency and reproducibility. To eliminate the dispersion effect, the empirical dispersion keyword = gd3 was added to the route section and the calculations were optimized without symmetry restrictions. MultiWFN code (Lu and Chen, 2012) was used to analyze the calculation results.

2.5. pKa calculations

Acid dissociation constant (pKa) was used to quantitatively measure the strength of an acid in solution (the low the pKa value, the stronger acidity the solvent) (Shokri et al., 2013). The pKa calculations were also performed using the Gaussian 16 package, and the geometry and harmonic frequencies of each species were optimized employing the

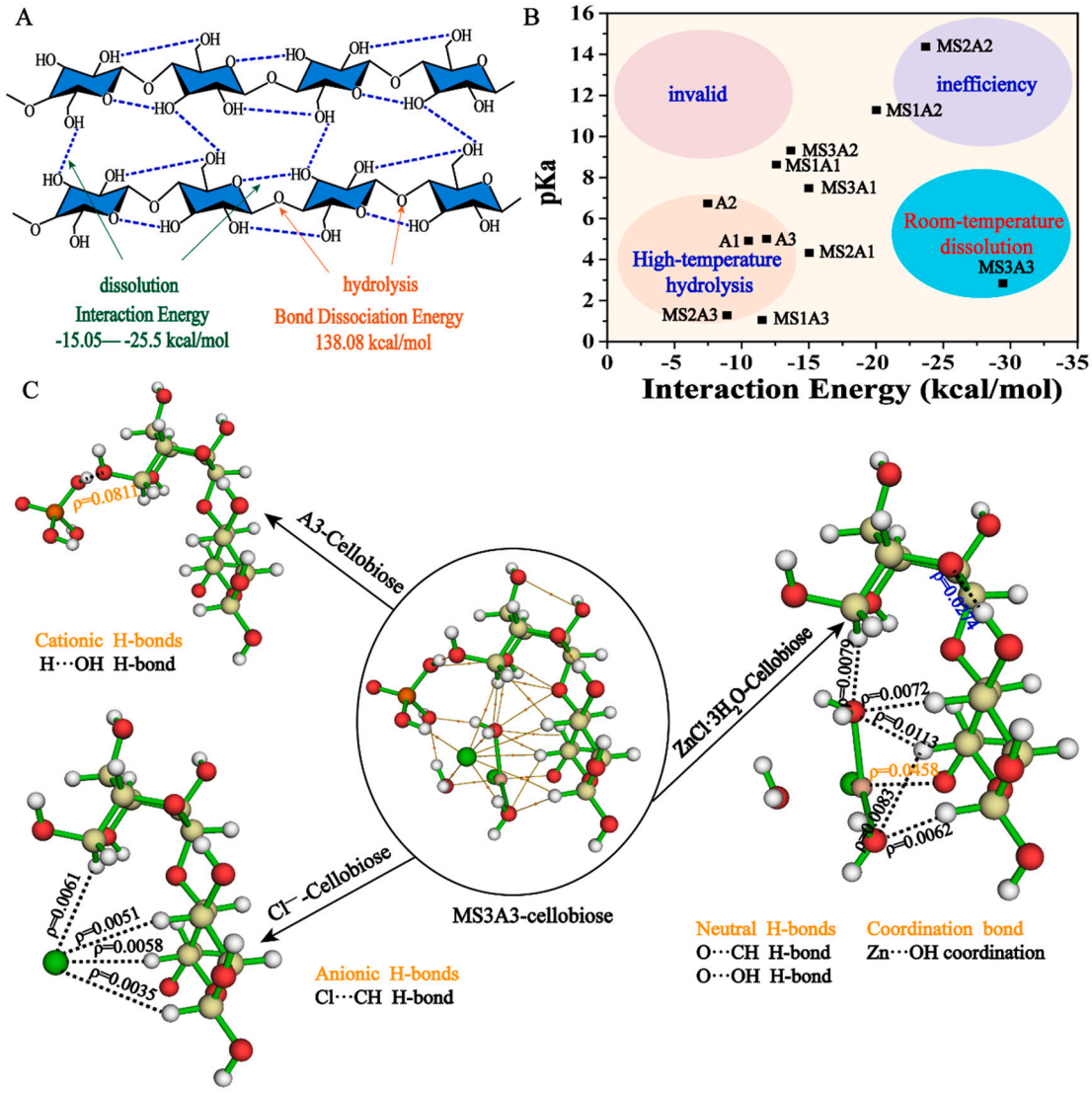


Fig. 1. The results of calculations for various combinations of solvent components to identify H-bond molecular scissors for cellulose. (A) The theoretical H-bond interaction energies between cellobiose units and the bond dissociation energy of glucosidic bond for cellobiose. (B) The calculated interaction energies and pKa values for combinations of metal salt hydrates (LiCl·3H_2O code-named as MS1, $MgCl_2$·3H_2O as MS2, and $ZnCl_2$·3H_2O as MS3) with different acids (oxalic acid as A1, acetic acid as A2, and phosphoric acid as A3). (C) Graphs of anionic, cationic and neutral H-bonds in the MS3A3-cellobiose system based on atom in molecules (AIM) theory.

B3LYP/6-311+G** method. Zero-point vibrational energy corrections were obtained by using the same method while single-point electronic energy values were calculated at the B2PLYPD3/def2TZVP and B3LYP/6-311+G** levels. The detailed calculation methods see Note S2.

2.6. Kamlet–Taft solvatochromic parameters

The Kamlet–Taft parameters (α, β, π^*) were determined from the absorption peaks of the following three dyes: Nile Red (NR), 4-nitrophenol (OH) and 4-nitroanisole (OMe). The detailed procedures are described as Note S3.

2.7. Degree of polymerization

The degree of polymerization (DP) of each cellulose specimen was determined according to the procedure in ISO 5351-2010, using an Ubbelohde viscometer (Xinbiao Tengda Instruments, Beijing, China) and copper(II) ethylenediamine as the solvent.

2.8. Performance liquid chromatography

High performance liquid chromatography (HPLC) data were acquired by completely dissolving each cellulose sample in the solvent and then precipitating the material by adding water and filtering the solution. The filtrate was neutralized with 0.1 M NaOH and the resulting zinc hydroxide precipitate was removed by filtration. The glucose and fructose concentrations in the filtrate were subsequently determined with an HPLC apparatus (LC-1260, Agilent, USA). This instrument incorporated a G7162A differential refractive index detector and a Bio-Rad Aminex HPX-87P column (300 mm × 7.7 mm) at 80 °C; injection volume 20 μL; ultrapure water as the mobile phase at a flow rate of 0.6 mL min^{-1}. Under these conditions, the retention times of glucose and fructose were 11.3 and 13.9 min, respectively.

2.9. Viscosity measurement

A DVS^+ digital viscometer (Brookfield Co. Ltd., Stoughton, MA) was used to measure the sample viscosity at room temperature (relative

Table 1
The electron density (ρ) and Laplacian ($^{\nabla 2}\rho$) values for the $ZnCl_2{\cdot}3H_2O$/phosphoric acid (MS3A3)-cellobiose system at the bond critical points as calculated at the M062X/6-311+G** level.

System	Interaction	Sort	Length (Å)	ρ (0.002–0.035)	$^{\nabla 2}\rho$ (0.024–0.139)
MS3A3-cellobiose	$Cl_{12}\cdots H_{43}$-C_{25}	$Cl^-\cdots C_6H$	3.132	0.0061	0.0194
	$Cl_{12}\cdots H_{53}$-C_{27}	$Cl^-\cdots C_3H$	3.102	0.0051	0.0149
	$Cl_{12}\cdots H_{49}$-C_{32}	$Cl^-\cdots C_1H$	3.378	0.0035	0.0094
	$Cl_{12}\cdots H_{54}$-C_{26}	$Cl^-\cdots C_5H$	3.095	0.0058	0.0165
	$H_{19}\cdots O_{55}$-H_{44}	**$H^+\cdots C_6OH$**	**1.458**	**0.0811**	**0.1382**
	$Zn_1\cdots O_{64}$-H_{39}	**$Zn^{2+}\cdots C_6OH$**	**2.214**	**0.0458**	**0.2021**
	$O_4\cdots H_{49}$-C_{32}	Water$\cdots C_1H$	2.705	0.0062	0.0238
	$O_4\cdots H_{54}$-C_{26}	Water$\cdots C_5H$	2.546	0.0083	0.0313
	$O_6\cdots H_{43}$-C_{25}	Water$\cdots C_6H$	2.602	0.0079	0.0303
	$O_6\cdots H_{54}$-C_{26}	Water$\cdots C_5H$	2.386	0.0113	0.0445
	$O_6\cdots H_{33}$-C_{30}	Water$\cdots C_6H$	2.758	0.0072	0.0298
	$O_{63}\cdots H_{34}$-O_{56}	Intra H-bond	1.917	0.0274	0.0867
MS3A3	$Cl_{12}\cdots H_{15}$-O_{14}	$Cl^-\cdots$P-OH	1.995	0.0386	0.0752
	$Zn_1\cdots O_{16}$-H_{17}	$Zn^{2+}\cdots$P-OH	2.872	0.0095	0.0454
Cellobiose	$O_{43}\cdots H_{14}$-O_{36}	Intra H-bond	1.906	0.0280	0.0872

humidity 34%) and with a speed of 12 rmp. Five replicates of each sample were tested to obtain the average value.

2.10. X-ray diffraction

The crystal phase of each cellulose sample was assessed by X-ray diffraction (XRD) with a D/max 2200 X-ray diffractometer (Rigaku, Tokyo, Japan) using Cu Kα radiation ($\lambda = 0.154$ nm) over the 2θ range from 5° to 40°. The crystallinity index was estimated using the Segal method (Segal et al., 1959).

2.11. Fourier transform infrared spectroscopy

Fourier transform infrared (FTIR) analyses were carried out using a Magna 560 spectrometer (Thermo Fisher Co., Waltham, MA) by acquiring spectra over the range of 650–4000 cm^{-1} at a resolution of 4 cm^{-1}. The characteristic peaks located at 3300 cm^{-1}, 1020 cm^{-1} and 895 cm^{-1} are assigned to O—H stretching, asymmetric stretching of glucose ring and vibrations of β-glucoside linkage between glucose units in cellulose (Wang et al., 2020).

2.12. Solid nuclear magnetic resonance spectroscopy

^{13}C nuclear magnetic resonance (NMR) analyses were performed using a 300 MHz AVANQIII spectrometer (Bruker, Germany) with a (CP/MAS) unit at room temperature. The spinning rate and the contact time were 8 kHz and 13.15 ms, respectively. The peaks located at 103 ppm, 79 ppm, 74 ppm and 60 ppm are assigned to the C1, C4, C2,3,5 and C6 carbon in cellulose, respectively (Cai and Zhang, 2005).

2.13. X-ray photoelectron spectroscopy

Ag 3d 5/2 X-ray photoelectron spectroscopy (XPS) data were obtained with an X-ray photoelectron spectrometer (Thermo Fisher Co., Waltham, MA) with a half peak width of 0.85 eV over the binding energy range 0–1200 eV, operating at 12 kV and 30 mA with an Al target.

2.14. Thermogravimetric analysis

A Pyris 6 thermogravimetric (TG) analyzer (PerkinElmer, Waltham, MA) was used to analyze the thermal stability of sample in a nitrogen atmosphere, applying a heating rate of 10 °C min^{-1}.

2.15. Scanning electron microscopy

Scanning electron microscopy (SEM) images of MCC and its regenerated product were taken by using a JSM-7500F microscope (JEOL, Japan) at 15 kV.

3. Results and discussion

3.1. Designing a H-bond molecular scissors solvent

The dissolution of cellulose depends on the free energy change in the mixed system, and this value is the sum of enthalpy and entropy changes. The driving force for dissolution is always an increase in the absolute value of entropy (Zhang et al., 2019). Entropy-driven dissolution reactions are of theoretical importance because free energy is the driving force for chemical changes. Therefore the interaction energy of cellobiose (a simple model for cellulose) was first analyzed by DFT calculations. The theoretical H-bonding interaction energy between cellobiose units was determined to be −15.05 kcal mol^{-1} (Supplementary Fig. S1), while the glucosidic bond dissociation energy of cellobiose was up to 138.08 kcal mol^{-1} (Fig. 1A). Ishida previously calculated the intermolecular interaction energies of cellulose at 25 °C, including the effects of reducing end groups, internal groups and non-reducing end groups in the gas phase as well as in water and ionic liquids, and obtained values in the range of −16.8 to −25.5 kcal mol^{-1} (Ishida, 2020). The criterion for selecting solvents was that the interaction energy between the solvent and cellobiose should be greater than that between the cellobiose units but should not exceed the glucosidic bond dissociation energy, so as to achieve an efficient dissolution and simultaneously avoid excessive hydrolysis (Cho et al., 2011). Considering the complexity of cellulose-solvent interactions, the lower limit of the interaction energy for efficiently dissolving cellulose should be higher than 25.5 kcal mol^{-1}.

Three metal salt hydrates were evaluated for the synthesis of new solvents by DFT analysis, including the $LiCl{\cdot}3H_2O$ (symbolized herein as MS1), $MgCl_2{\cdot}3H_2O$ (MS2) and $ZnCl_2{\cdot}3H_2O$ (MS3). Three acids were also assessed: oxalic acid (A1), acetic acid (A2) and phosphoric acid (A3). The interaction energies of the combinations of these compounds were calculated by DFT (Supplementary Fig. S2) and are summarized in Fig. 1B, from which it is evident that only the H-bonding interaction energy of the MS3A3 pair met the criterion of being larger than 25.5 kcal mol^{-1}. The interaction energies between each solvent combination and cellobiose were calculated based on DFT (Supplementary Fig. S3) and are provided in Supplementary Table S1. The MS3A3 system had the highest interaction energy with cellobiose (−29.44 kcal mol^{-1}), indicating that it could be the most effective solvent according to the criterion of selecting solvent (Supplementary Fig. S4). The experimental observation on cellulose dissolution also confirmed that only the MS3A3 system was capable of effectively dissolving cellulose (Supplementary

Table 2
Properties of four types of cellulose feedstocks.

Parameters	Microcrystalline cellulose	Wood cellulose	Pulp cellulose	Cotton linters
Degree of polymerization	183	717	1539	2243
Average molecular weight	2.96×10^4	1.16×10^5	2.49×10^5	3.63×10^5
Content of α-cellulose	98.31%	95.75%	93.90%	99.97%
Content of hemicellulose	1.73%	3.89%	5.90%	–
Content of lignin	–	0.36%	0.20%	0.03%
Crystallinity index	62.12%	62.41%	53.58%	60.72%

Fig. S5). In addition, the pKa value of the MS3A3 (Supplementary Fig. S6) is significantly lower than those of the majority of the other combinations and of any of the pure acids (Supplementary Table S2). This result suggests that MS3A3 has more numerous dissociated hydrogen ions, facilitating to improve the ability of solvent to destroy intra- and intermolecular hydrogen bonds within cellulose. Other combinations of metal salt hydrates and acids are deemed to be either inefficient or only fit for high-temperature hydrolysis of cellulose. On this basis, the MS3A3 system was considered an ideal solvent to test the aforementioned hypothesis.

The atoms in molecules (AIM) theory is often employed to quantify the H-bond interactions at bond critical point (BCPs) in terms of electron density (ρ, 0.002–0.035 au) and the Laplacian of the electron density ($^{\nabla 2}\rho$, 0.024–0.139 au). The ρ value indicates the bond strength while the $^{\nabla 2}\rho$ value indicates the bond type (that is, H-bond, ionic bond or van der Waals force). In our solvent, the hydrogen atom of hydroxyl group in phosphoric acid molecule can form H-bonds with Cl^- ion ($\rho = 0.0386$ au; $^{\nabla 2}\rho = 0.0752$ au) with a distance of 1.995 Å (Table 1 and Supplementary Fig. S4). This promotes the release of Cl^- ions from the Zn^{2+} ions, such that the distance between Zn^{2+} and Cl^- ions is increased from 2.464 to 2.801 Å. Simultaneously, the Zn^{2+} ions coordinate with the hydroxyl groups of the phosphoric acid. The resulting interatomic distances are within the range of less than 3.5 Å over which H-bonding can occur.

Fig. 1C and Supplementary Fig. S4 exhibit the topological graph of the MS3A3-cellobiose system by AIM analysis. Table 1 summarizes the ρ and $^{\nabla 2}\rho$ values for all BCPs between MS3A3 and cellobiose. The calculation results indicate that H^+ ions are released from the phosphoric acid that forms strong H-bonds with O atoms in C_6-OH groups ($\rho = 0.0811$ au, $^{\nabla 2}\rho = 0.1382$ au). Meanwhile, the distance between Zn^{2+} and Cl^- ions also increases to 3.793 Å, indicating that the Cl^- ions are no longer tightly bonded to the Zn^{2+} ions, thus, these Cl^- ions subsequently form H-bonding interactions with C_1-H groups in the cellobiose as well as C_3-H, C_5-H and C_6-H moieties, leading to selectively compete or wrap H-bond of cellulose itself. It also adjusts the amphiphilicity of the solvent, thereby increasing the ability to dissolve cellulose. At the same time, Zn^{2+} ions form coordination bonds with the oxygen atoms in C_6-OH groups in the cellobiose ($\rho = 0.0458$ au, $^{\nabla 2}\rho = 0.2021$ au) and water molecules form H-bonds with the cellobiose, which also facilitate the cellulose to dissolve. In addition, the ρ and $^{\nabla 2}\rho$ values in the cellobiose itself both decrease, indicating that the MS3A3 is theoretically capable of dissolving cellulose and is superior to MS3 and A3 alone (Supplementary Table S3).

3.2. Room temperature dissolution of cellulose using MS3A3

The resulting MS3A3 solvent system has several advantages. Specifically, $ZnCl_2$ is readily available and inexpensive. In addition, phosphoric acid is a weak oxidizer and has been previously used as an economical and effective solvent for high molecular weight cellulose. Given strong H-bond interaction energy, the acidity of solvent, the coordination effect of zinc ions, and the ability to adjust the amphiphilicity of chloride ions, it is evident that the MS3A3 system has the potential to

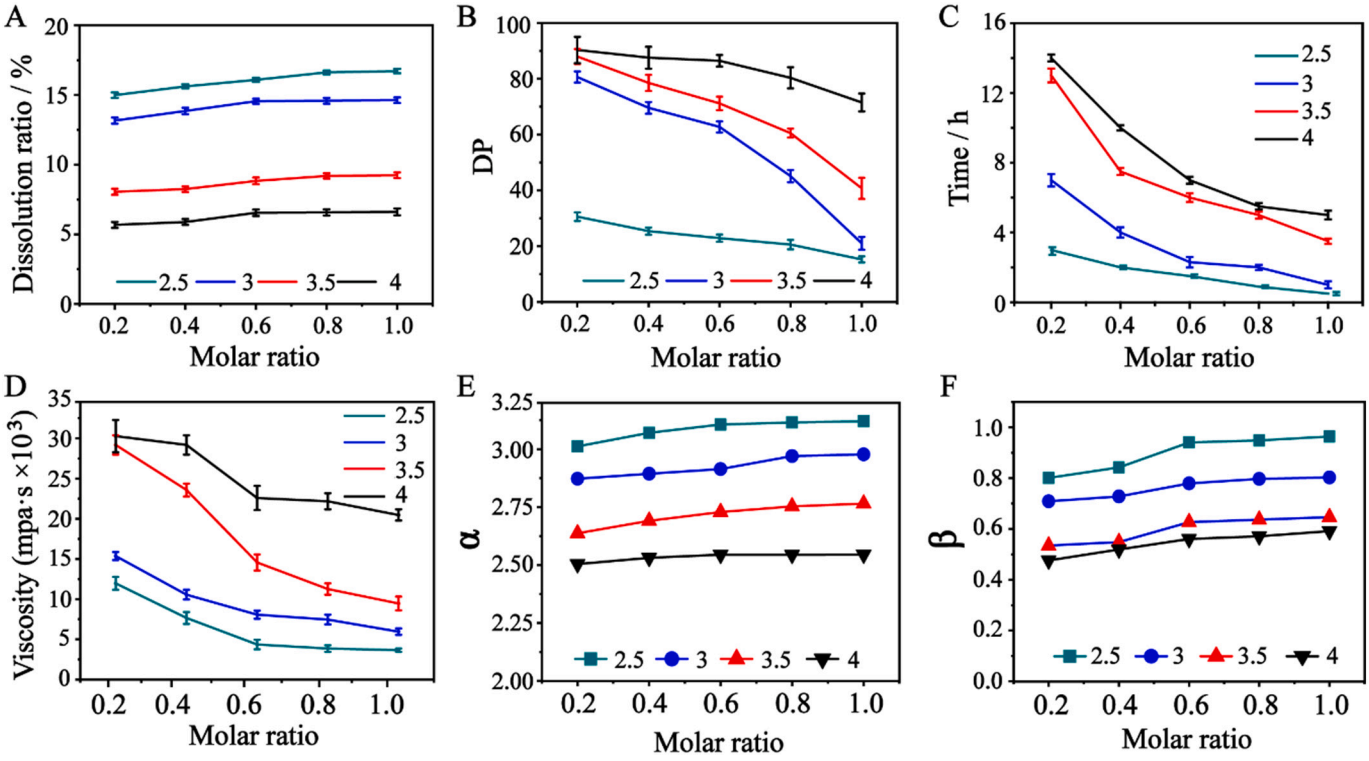


Fig. 2. The dissolution of cellulose using $ZnCl_2{\cdot}3H_2O$/phosphoric acid (MS3A3) solvents. (A) The dissolution ratios of microcrystalline cellulose (MCC) at room temperature, (B) the degree of polymerization (DP) values of the recovered MCC, (C) the dissolution time, and (D) the viscosity of MCC solutions, as functions of the molar ratios of water (2.5, 3, 3.5 or 4) and H_3PO_4 (0.2, 0.4, 0.6, 0.8 or 1.0). (E) α and (F) β values of the MS3A3 solvent with variations in the component molar ratios.

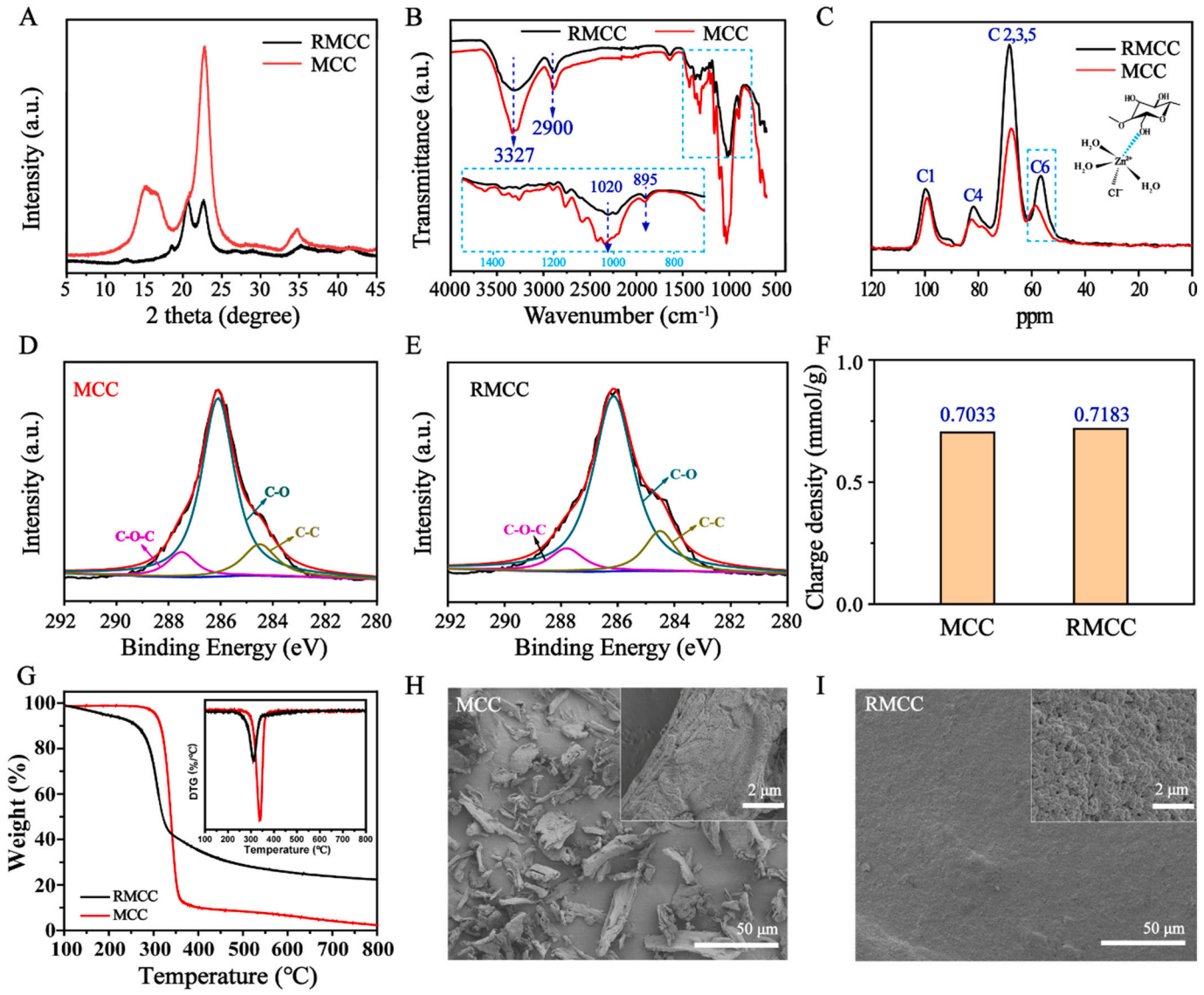


Fig. 3. The structure and properties of cellulose recovered from solution. (A) X-ray diffraction (XRD), (B) Fourier transform infrared (FTIR) and (C) ^{13}C solid nuclear magnetic resonance spectra of the microcrystalline cellulose (MCC) and regenerated MCC (RMCC). XPS spectra showing surface element distributions and XPS C 1s high-resolution spectra of the (D) MCC and (E) RMCC. (F) Charge density of MCC and RMCC. (G) Thermogravimetry (TG) and differential thermogravimetry (DTG) data obtained from the MCC and RMCC. Scanning electron microscopy (SEM) images of the (H) MCC and (I) RMCC (magnification, scale bar: ×1000, 50 μm and ×20,000, 2 μm).

dissolve cellulose at room temperature. To verify this, we synthesized a series of these solvents with the components added at varying molar ratios. Four types of cellulose were selected as the feedstocks for the dissolution experiments (Table 2). In these trials, microcrystalline cellulose (MCC) was firstly used as a model cellulose to ascertain the optimal solvent composition ratio and dissolution conditions. The resulting dissolution data along with dissolution values, DP, viscosity and time are summarized in Fig. 2.

Fig. 2A and Supplementary Fig. S7 demonstrate that the extent of cellulose dissolution was greatly affected by the water content of the solvent, such that a lower proportion of water gave increased dissolution. However, the incorporation of less water in the solvent (i.e., 2.5) made the system excessively acidic to the extent that the DP value of cellulose decreased (Fig. 2B). Small molecules such as glucose and fructose were occasionally found (Supplementary Table S4), suggesting that a portion of the MCC was hydrolyzed into saccharides. On the basis of these data, the optimal molar ratio of water in the solvent was determined to be 3. With the increases in the proportion of phosphoric acid, the solubility of the MCC almost unchanged, while the dissolution time and the viscosity of the solution both decreased slightly (Fig. 2C and D). An inflection point associated with an optimal balance of the various parameters was observed at a phosphoric acid molar ratio of 0.6. Based on the above analysis, the optimal MS3A3 composition was determined to be $ZnCl_2$:H_2O:H_3PO_4 as 1:3:0.6 (i.e. MSA-3-0.6). The viscosity of the MSA-3-0.6 ranged from 40 to 50 mPa·s and, using this solvent, the MCC was dissolved at room temperature with a maximum dissolution of 15 wt%.

The Kamlet-Taft (K-T) solvatochromic method was also used to estimate the properties of the solvents (Supplementary Fig. S8) to disclose their relations with the dissolution of cellulose. In the K-T method, the parameter α represents the H-bond acidity of the solvent, β indicates the H-bond basicity and π^* describes the dipolarity/polarizability (Jessop et al., 2012). The α values of pure A3 and of the MS3A3 system were determined to be 2.69 and 2.91, respectively (Supplementary Table S5). These results suggest that the addition of the metal salt promoted the release of H^+ ions, which was consistent with the results of the DFT

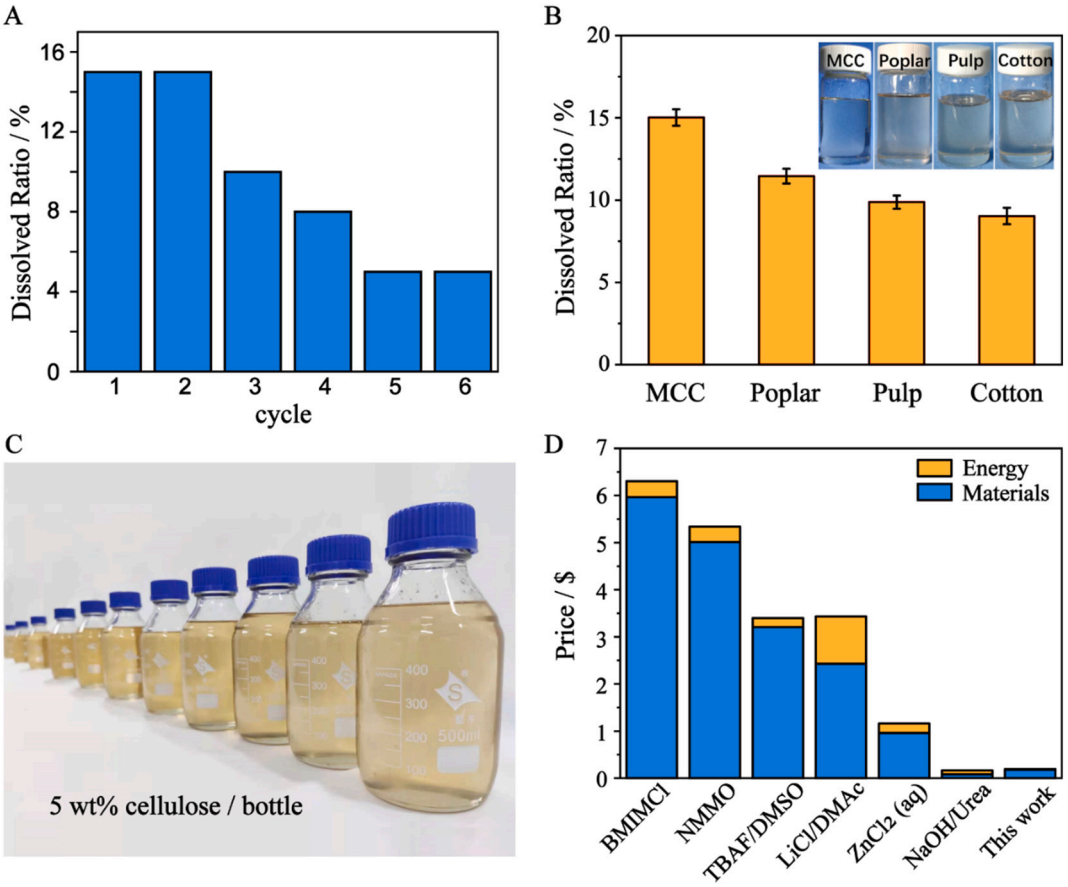


Fig. 4. Recyclability, universality, and low-cost dissolution of cellulose using the MSA-3-0.6 (zinc chloride, water and phosphoric acid mixed at molar ratios of 1:3:0.6) solvent. (A) Dissolved ability of the MSA-3-0.6 during recycling uses for cellulose dissolution at room temperature (RT). (B) The maximum dissolution ratios of four types of cellulose using the MSA-3-0.6. (C) Photographic images of 5 wt% cellulose dissolved in the MSA-3-0.6 from a pilot production experiment. (D) The cost of raw materials of solvent consumed for dissolving 1 g of cellulose with different solvents and the energy consumption during the dissolution process.

calculations. Simultaneously, the original β values of the pure MS3 and A3 were found to be 0.52 and 0.40, while that of the MS3A3 was increased to 0.78. It is possible that the Zn^{2+} cations and Cl^{-} anion increased the H-bonding acceptability of the solvent and so were conducive to the dissolution of cellulose. The α and β values of the MSA-*x*-*y* solvent series showed that the molar ratio of water had the greatest effect on these values (Fig. 2E and F). The variations in α and β values with the solvent composition were observed to be consistent with the changes in the extent of cellulose dissolution.

3.3. Structure and properties of the regenerated cellulose

The effects of this solvent on the properties of cellulose were examined by characterizing specimens after dissolution and recovery. XRD patterns were acquired to study the crystalline structure of the MCC and its regenerated product (RMCC). The patterns in Fig. 3A exhibited diffraction peaks at $2\theta = 15.5^{\circ}$ and 22.7° that correspond to type I crystal structure. The emergence of new peaks at $2\theta = 20.2^{\circ}$ and 22.2° in the RMCC pattern indicates a change to type II crystal phase as a result of the reconstruction of H-bonds during the recovery process. Similar results were also obtained with other types of cellulose (Supplementary Fig. S9).

We also investigated the Fourier transform infrared (FTIR) spectra of the MCC and RMCC to further investigate the chemical evolution during the dissolution process (Fig. 3B). The FTIR spectrum of the RMCC had the characteristic peaks related to O-H stretching at 3300 cm^{-1}, asymmetric stretching of the glucose ring at 1020 cm^{-1} and vibrations of β-glucoside linkages (C_1-O-C_4) between glucose units at 895 cm^{-1}, indicating that the cellulose composition was essentially unchanged. It is worth noting that these spectra do not show peaks at 829 and 927 cm^{-1} corresponding to P-O-C and P-OH groups, and that the XPS data for the RMCC did not exhibit a P 2p peak. ^{13}C solid NMR spectroscopy was used to ascertain changes in the chemical shift of the MCC and RMCC (Supplementary Table S6). As shown in Fig. 3C, the C2,3,5 and C6 signals were very intense, which might be caused by the structure change of cellulose. After cellulose regeneration, the loose cellulose powder became denser. The chemical shift for signal peak of C6 was extremely obvious, demonstrating that the MS3A3 primarily interacted with the C6-OH groups of the cellulose, which is consistent with the above theoretical calculation and analysis results. The C 1s XPS data obtained from the MCC and RMCC are shown in Fig. 3D and E, and both of them have typical types of carbon bonds: C-C at 284.5 eV, C-O at 286.1 eV and O-C-O at 287.5 eV, indicating that cellulose retains its complete structure. In addition, the charge densities of the MCC and RMCC were also almost equivalent, at 0.7033 and 0.7183 mmol g^{-1} (Fig. 3F and Supplementary Fig. S10). The results of FTIR, XPS and charge densities all confirm that phosphorylation of the cellulose did not occur during the dissolution process. Therefore, the most challenging aspect of dissolving cellulose (that of finding a non-derivatizing solvent that does not react with the material) was successfully achieved.

The TG data of the MCC and RMCC indicated that the thermal stability of the RMCC (310 °C) was lower than that of the MCC (~340 °C) and that the residual carbon mass of RMCC (22.49%) was much more than that of MCC (Fig. 3G). Fig. 3H and I presents SEM images of the MCC and RMCC, showing that the RMCC with a flat, more refined micromorphology was generated from the dissolution and recovery

process.

3.4. Demonstrations of pilot production and recyclable dissolution

In the process of industrial production of cellulose dissolution, the desired solvent must have the characteristics of recyclability, good universality, pilot production, high efficiency, and low cost. In the solvent system that we designed, the MS3A3 demonstrates excellent recyclability. Even after recycling, the MS3A3 maintained its high ability to dissolve cellulose (original solvent: 15 wt%; after 1 time: 15 wt%, Fig. 4A and Supplementary Fig. S11). After reusing the MS3A3 for five times, the dissolved ratio of cellulose was 5 wt%, indicating the effect of the H-bond molecular scissors of the MS3A3 was retained even after being recycled and reused. In addition, the maximum dissolution ratios of other cellulose feedstocks with different molecular weights were also assessed (Fig. 4B), and in each case the extent of dissolution was considered satisfactory (Supplementary Fig. S12). Although the DP of cellulose feedstock increased from 183 to 2243, the dissolution time and the viscosity of solution did not change greatly (Supplementary Table S7), which suggests the broad applicability of MS3A3 for dissolving various types of cellulose. Fig. 4C presents photographic images of a pilot production experiment of dissolving 550 g of MCC with the MSA-3-0.6 at room temperature, demonstrating its suitability not only in the laboratory uses but also at the pilot scale. The cost effectiveness and minimal energy consumption associated with dissolution using the MSA-3-0.6 solvent indicates that this system could compete with the NaOH/urea system and superior to many other solvents (Fig. 4D and Supplementary Table S8). All these results support that the new solvent has a very broad and quite optimistic application prospects.

4. Conclusion

In summary, with the assistance of DFT calculations, a new solvent comprising zinc chloride, water and phosphoric acid (at molar ratio of 1:3:0.6) was successfully designed for the room temperature dissolution of cellulose. The data show that $ZnCl_2{\cdot}3H_2O$ and H_3PO_4 produce a new DES with strong H-bond donor-acceptor properties in which with water acts as an intermediate mediator to provide polar environment and avoid cellulose hydrolysis. This solvent has a protonation property and a characteristic of both having strong H-bonding acidity and basicity to achieve in efficiently dissolving cellulose (15 wt%). The newly developed solvent also has excellent universality, which can successfully dissolve the cellulose from various sources (e.g., microcrystalline, pulp cellulose, wood cellulose and cotton linter) at room temperature. This dissolution process did not phosphorylate the cellulose, although the cellulose recovered from solution showed a change in crystal structure from type I to II. In summary, this tailored solvent has a broad and quite optimistic application prospects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the China National Science Fund for Distinguished Young Scientists (Grant No. 31925028), and the National Undergraduate Training Programs for Innovations (Grant No. 201910225192).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbpol.2021.118473.

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